

M-REVS: Multimodal Real Estate Valuation System

Overview

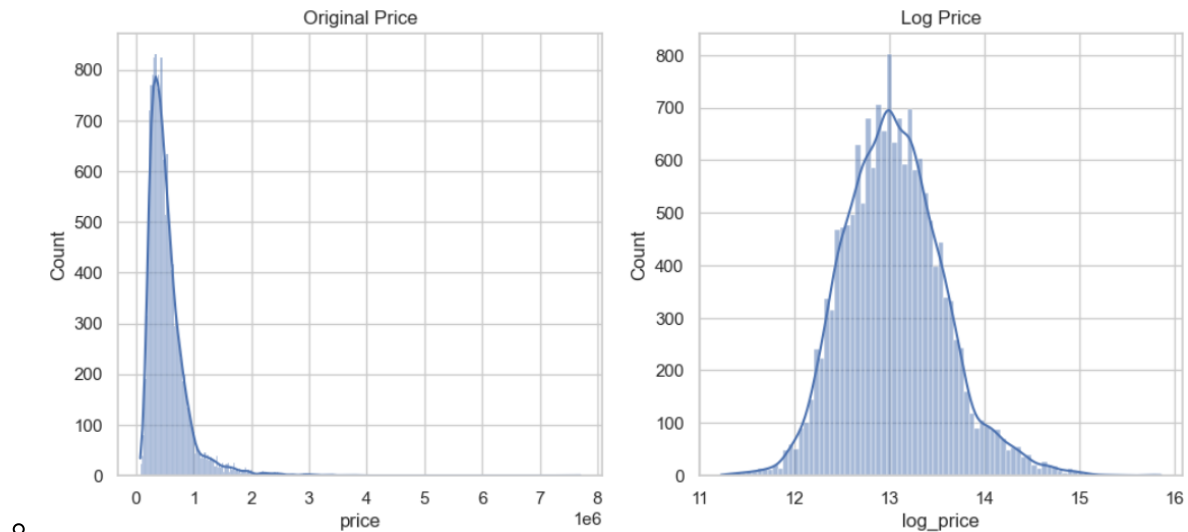
M-REVS fuses structured housing attributes with satellite imagery to mitigate omitted-variable bias in Automated Valuation Models (AVMs) for King County. The pipeline combines gradient-boosted decision trees (XGBoost, LightGBM), a shallow neural head for tabular signals, and frozen ResNet50 embeddings for visual cues, unified via a stacking meta-learner. Tooling spans PyTorch (vision/fusion), scikit-learn (preprocessing and meta-learner), and XGBoost/LightGBM (tabular experts). Compared to a strong tabular-only baseline (R^2 0.8888, RMSE $\approx$ \$113k), the multimodal stack lifts accuracy to R^2 0.9388 with MAE $\approx$ \$52k.

Data and Preprocessing

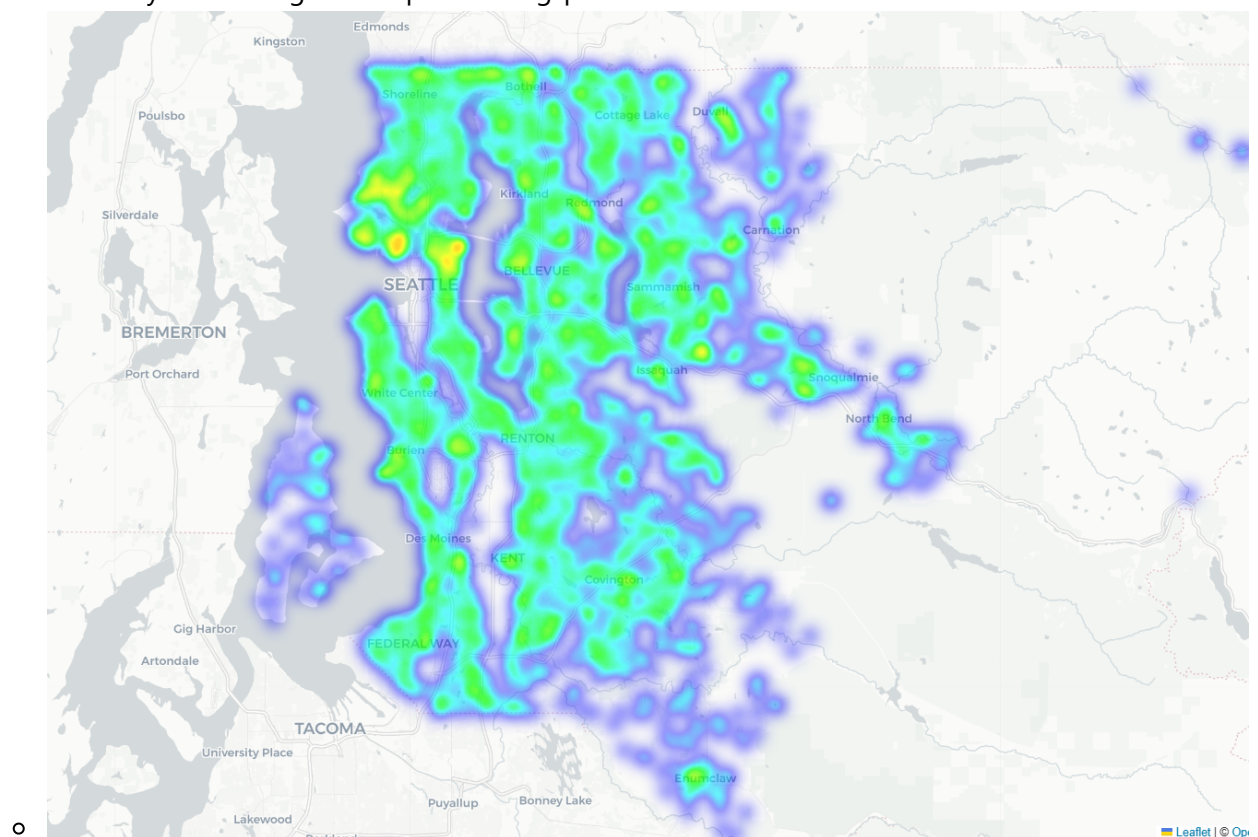
- **Sources:** King County housing records with lat/long matched to satellite tiles.
- **Cleaning:** Enforced `sqft_living = sqft_above + sqft_basement`; removed rows with bedrooms ≤ 0 or `sqft_living < 200`.
- **Target transform:** `log1p(price)` reduces skew from 4.03 to 0.41, improving loss smoothness and residual normality.
- **Spatial features:** K-Means ($k=20$) on coordinates yields `cluster_id`; geodesic distance to downtown Seattle captures access premium. K-Means persisted (`kmeans_model.pkl`) for inference stability.
- **Feature engineering:** `lot_to_living_ratio`, `sqft_per_room`, `age = 2024 - yr_built`, plus legacy size/context (`sqft_living15`, `sqft_lot15`, `dist_to_downtown`).
- **Scaling/encoding:** StandardScaler on continuous/coordinate subsets (fit on train only); LabelEncoder for clusters in fusion track; TargetEncoder + StandardScaler in baseline track.
- **Visual embeddings:** ResNet50 (pretrained, frozen) with global average pooling $\rightarrow$ 512-d vectors stored in `image_features.npy`; computed once to decouple vision from tabular training.

Exploratory Data Analysis (EDA)

- **Price distribution:** Raw prices are right-skewed; `log1p` delivers near-symmetric residuals, improving booster stability.
- **Spatial heterogeneity:** `cluster_id` separates waterfront/urban cores from inland suburbs; distance-to-downtown shows monotonic price decay.
- **Visuals:**



– side-by-side histograms of price vs. log-price.



– folium/heatmap of prices across King County highlighting waterfront and downtown gradients.

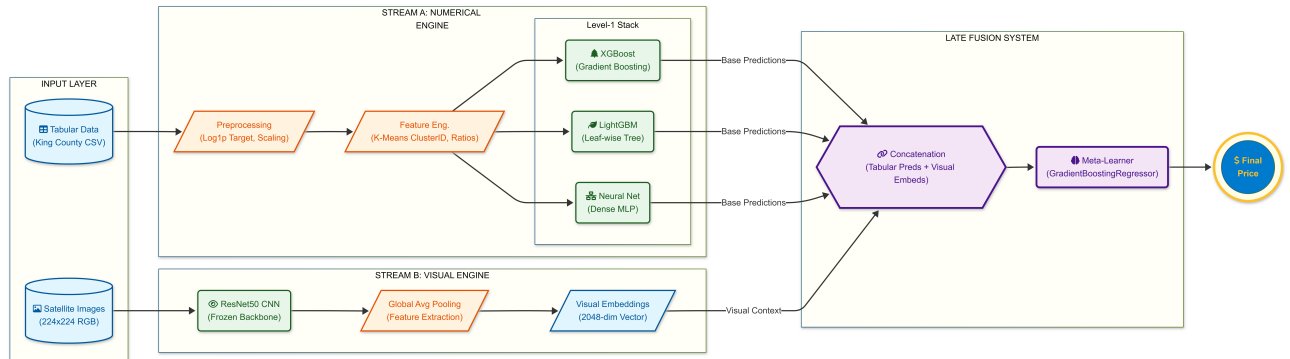
Architecture

- **Stream A — Tabular experts:**

- XGBoost (stack track): ~3000 trees, $lr \approx 0.01$, $max_depth \approx 8$, $subsample \approx 0.7$, $colsample_bytree \approx 0.6$, early stopping.
- LightGBM (leaf-wise): 3000 trees, $lr = 0.005$, $num_leaves = 45$, $feature_fraction = 0.6$, $bagging_fraction = 0.7$.
- Shallow MLP head (ResidualFusionNet tab branch): Linear $\rightarrow$ BatchNorm $\rightarrow$ GELU $\rightarrow$ Dropout for structured inputs with cluster embedding.

- **Stream B — Visual encoder:** ResNet50 pretrained, frozen; global average pooling to 512-d embeddings; no fine-tuning to keep training lightweight and avoid overfitting.

- **Fusion:** Concatenate standardized tabular features (with cluster embedding) and 512-d visual embeddings within the fusion net; parallel expert predictions retained for stacking.
- **Meta-learner with disagreement features:** GradientBoostingRegressor ($n_estimators=1000$, $lr=0.01$, $max_depth=4$, $subsample=0.65$) consumes expert outputs plus pairwise deltas (XGB-NN, LGB-NN) to arbitrate conflicts and reduce over-reliance on any single modality.
- **Diagram:**



– Satellite Image → ResNet50 → 512-d embedding; Tabular Data → XGBoost/LightGBM/MLP → predictions; concat preds + disagreement → Meta-Learner → Final Price.

Training & Validation Workflow

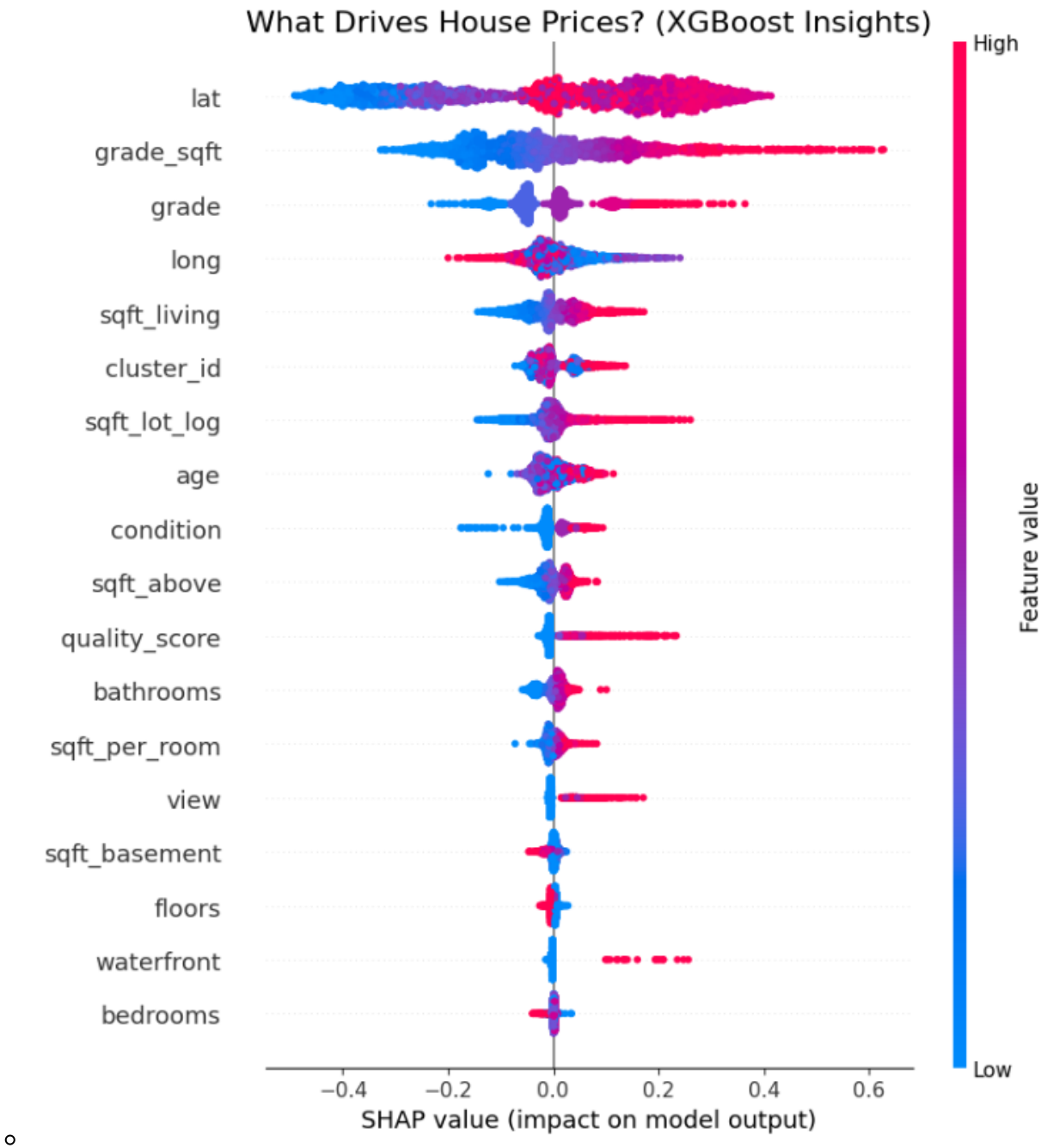
- Train/validation split performed before scaling to avoid leakage.
- StandardScaler fit on train indices only; LabelEncoder fit on full cluster set (safe for labels).
- Vision embeddings precomputed; loaders assemble (cont, cat, vis) tensors; optimizer AdamW with OneCycleLR; Huber loss for robustness in fusion net.
- Early stopping on boosters; best fusion checkpoint persisted as `best_fusion_model.pth`; stacking artifacts saved (`xgb_final.json`, `lgb_final.pkl`, `meta_learner_final.pkl`, `scaler_stack.pkl`).

Results

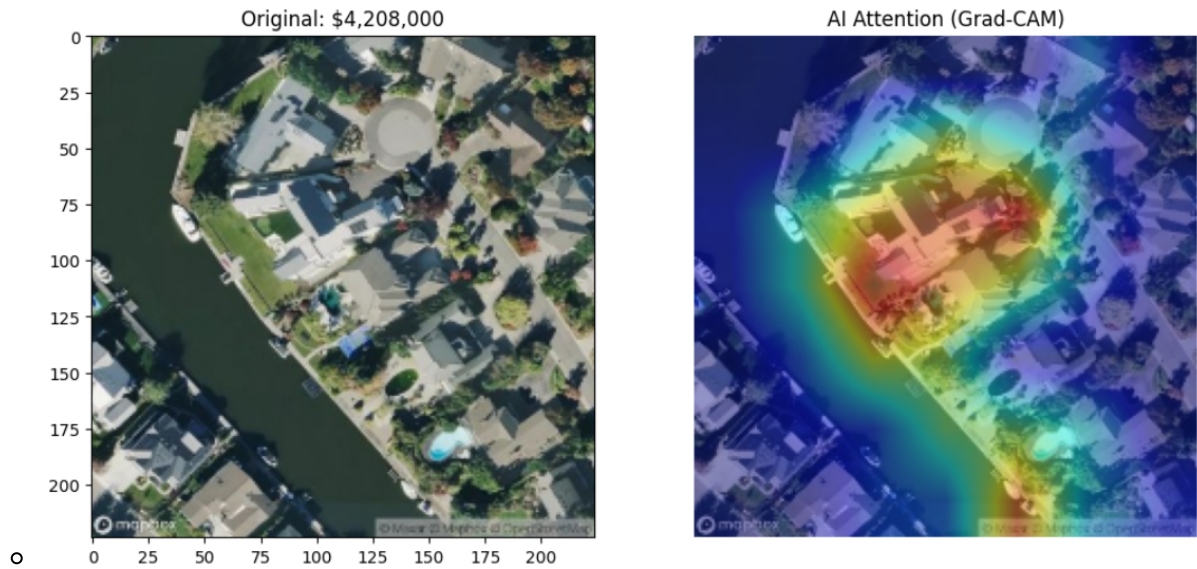
- **Tabular-only baseline (preprocessing track):** $R^2 = 0.8888$; $MAE \approx \$62,549$; $RMSE \approx \$113,523$ (XGBoost, 1000 trees, $lr=0.05$, $depth=6$, $subsample/colsample=0.8$).
- **Multimodal stacked champion (fusion/stack track):** $R^2 = 0.9388$; $MAE \approx \$52,001$; RMSE not logged (expected < baseline).
- Gains concentrate in visually distinctive segments (waterfront, high greenery, dense urban blocks) where tabular proxies under-represent aesthetics.

Financial and Visual Insights

- **SHAP (tabular experts):** `grade`, `sqft_living`, `sqft_lot`, `waterfront/view`, `cluster_id`, `dist_to_downtown` dominate importance; ratios (`lot_to_living_ratio`, `sqft_per_room`) capture efficiency; age/renovation capture depreciation.



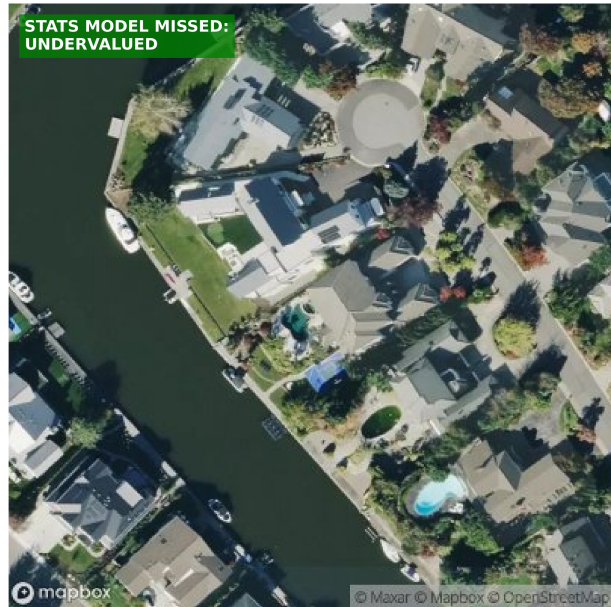
- beeswarm/bar view of top contributors.
- **Grad-CAM (visual stream):** Highlights tree canopy, water adjacency, and block openness for high-value predictions; extensive hardscape/industrial adjacency suppresses value. Visual attributions align with performance uplift, confirming imagery mitigates omitted visual bias.



– overlays on satellite tiles for high vs. low valuations.

Case Study: Visual Correction

Property ID: 6065300370



AI VALUATION AUDIT

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PROPERTY SPECS: 5 Bed | 6.0 Bath | 7440 SqFt | Grade 12

1. STATISTICAL BASELINE (XGBoost + LightGBM)
Prediction: \$2,446,851
(Based on numbers only. Lacks visual context.)

2. FINAL ENSEMBLE (Corrected by Neural Net)
Final Valuation: \$3,625,806
Adjustment: +\$1,178,955 (Visual Correction)

ACTUAL SOLD PRICE: \$4,208,000

ACCURACY ANALYSIS:
Stats-Only Error: \$1,761,149 (Missed by wide margin)
Final Model Error: \$582,194 (Close prediction)

SAVED ERROR: \$1,178,955
(The AI 'saw' value that the spreadsheet missed)

Baseline (tabular-only) undervalued a waterfront property; multimodal ensemble adjusted upward by ~\$1.18M, cutting error from ~\$1.76M to ~\$0.58M.

Business Impact

- Reduced typical error (MAE) by ~\$10.5k over a strong tabular baseline, improving confidence in underwriting, lending, and acquisition filters.
- Better calibration in visually premium micro-markets lowers false negatives on high-value assets and mitigates overpricing risk in visually disadvantaged areas.
- Modular design (frozen embeddings, persisted clustering) simplifies deployment and retraining cycles.

Recommendations & Future Work

- Log RMSE for the stacked model and add calibration (e.g., isotonic or Platt) for downstream risk usage.
- Periodically refresh satellite embeddings; consider light fine-tuning on real-estate tiles.
- Add conformal prediction intervals for decision risk bounds.
- Explore geospatial GNNs or attention over parcels to capture relational effects; incorporate temporal drift monitoring for clusters and pricing dynamics.

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